Smart Bed Monitoring for Patient Safety: Using Piezoelectric Sensors and ThingsBoard for Real-Time Care

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Abstract

Caring for patients who have a time moving around is a big problem for hospitals. Nurses have a time keeping an eye on every single bed all the time, which can lead to accidents like falls or sores on the skin.In this project we made a system to watch over patients. This system uses sensors and an ESP32 to see how patients are moving without bothering them.We tried out our idea using a Wokwi simulation. Each sensor takes in information about pressure. Turns it into detailed data.This helped us figure out a line to know when a patient is in bed or when they are moving around a lot.We linked our system to the ThingsBoard platform. This made a dashboard that shows the pressure now and what the patient was doing before.Our tests showed that if we check on the patient every 5 seconds we can give nurses an reliable tool to help keep patients safe.We think this tool can really help nurses take care of patients who cannot move easily.The patient movement system we made is a way to watch over patients without being in their face all the time.This is good, for patients who need a lot of care and for nurses who want to do their job.

Keywords: IoT, Smart Bed, Patient Monitoring, Piezoelectric Sensors, ThingsBoard, ESP32, Healthcare Technology.

1. Introduction

We all know that hospital staff have a time keeping an eye on every patient at all times. Patients who are stuck in bed can get hurt easily if they try to move or get up without any help. Many systems try to help hospital staff with this problem. These systems often use cameras or wearable sensors that patients find really annoying.In our project we wanted to create something that works quietly in the background without bothering patients. We picked piezoelectric sensors because they are really thin and can detect changes in pressure. From what we have seen in our research piezoelectric sensors are being used for lots of things like tracking heartbeats and detecting when someone gets out of bed.Piezoelectric sensors are really sensitive to pressure they can detect changes in pressure. Piezoelectric sensors are being used a lot for healthcare applications like tracking patient movement. [3, 8, 13, 15, 16]. Our goal was to take this piezoelectric sensor technology and connect it to a platform like ThingsBoard. This way healthcare providers can look at data that makes sense to them the healthcare providers. We want to make it easy for healthcare providers to understand the data, from this piezoelectric sensor technology. [1, 2, 6, 18, 20].

2. Building the Prototype

2.1 The Hardware Setup

Our system is built around the ESP32 microcontroller, which is widely used in IoT healthcare frameworks [5, 9, 12, 17]. We used three sensor modules, the ESP32 microcontroller and these sensors worked well together.We placed the sensor modules at the Left, Center and Right zones of the bed.The ESP32 microcontroller was very important here.We wanted to see how intense the pressure really was on the bed.So we did not just want a yes or no signal from the ESP32 microcontroller.Instead we made the simulation give us analog values from the ESP32 microcontroller. (available at `https://wokwi.com/projects/458672942543016961`) give us analog values. This gave us a range from 0 to 4095 on the ESP32s 12-bit ADC. Now we could see small changes in weight. The ESP32 microcontroller helped us do that.

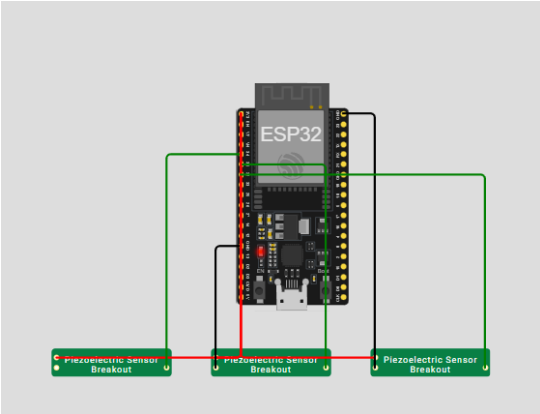


Figure 1: The Wokwi simulation layout where we tested the ESP32 and sensor wiring.

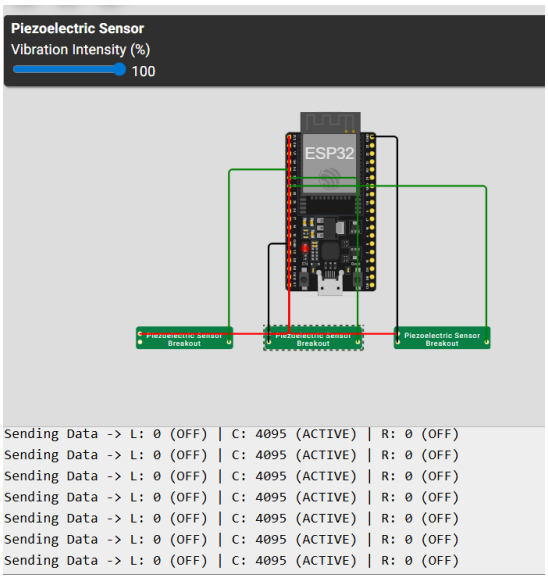
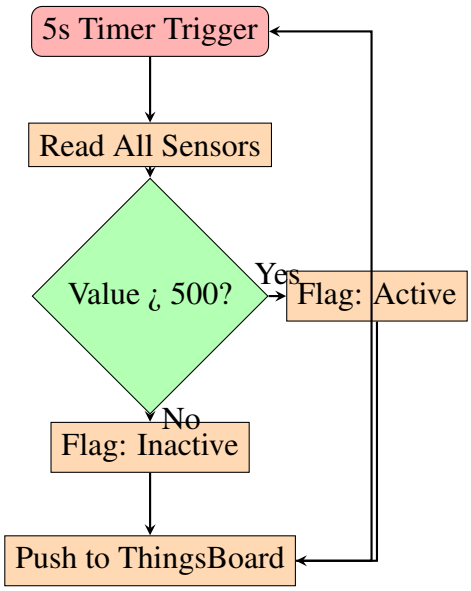


Figure 2: Serial Monitor output showing the local processing of sensor data segments.

2.2 The Software Logic

The ESP32 code is quite easy. It does the job. The ESP32 reads the sensors. Checks the values against a certain limit. When we tested the ESP32 we figured out that a value of 500 is just right. This value is low enough to know when someone is lying down. It is high enough to ignore tiny movements or other noise. The ESP32 collects all the data, from the sensors every 5 seconds. Sends the data to the cloud.



3. Observations and Results

When we tested the system it was really cool to see how well the sensors caught movement. The sensors worked great. They picked up movement well. It was actually motivating.

3.1 Live Monitoring

We created gauges, in the ThingsBoard dashboard. It makes it really easy for a nurse to look at a screen and know if a patient is leaning to one side.. If they are centered.

Figure 3: These gauges show the live pressure levels from the three sensors.

3.2 Pattern Tracking

The historical graphs were really helpful in this project. When I looked at the spikes in the graph I could see when a patient was restless at night. It was easy to track their periods.The graphs, in Fig. 5, Showed this information clearly

Figure 4: Complete dashboard layout uniting real-time gauges with telemetry status.

Figure 5: The history chart lets us see exactly when movements happened over time.

4. Lessons Learned and Future Goals

This project showed us that we do not need to buy equipment to make a good monitoring tool. The simple equipment we are using now is good enough to detect what we want. We have some ideas we want to try with the monitoring tool. We believe that if we use some machine learning ideas with the monitoring tool the monitoring tool will be able to detect how a patient is lying in bed or even figure out how fast they are breathing just by feeling the vibrations in the mattress. [4, 7, 10, 11, 14, 19, 21]. We are talking about the monitoring tool and the same machine learning ideas that we think will help us make the monitoring tool better. The mattress will be a part of the monitoring tool because it is where the patient will be lying and where we will be measuring the vibrations, with the monitoring tool.

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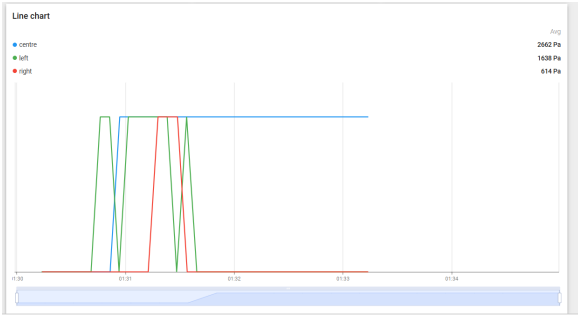

Figure 6: Close-up view of overlapping sensor signals indicating a patient's shift in position.

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